AUREX INTERNSHIP

FULL-STACK ENGINEERING — MONTH 3 INDUSTRY PROJECTS

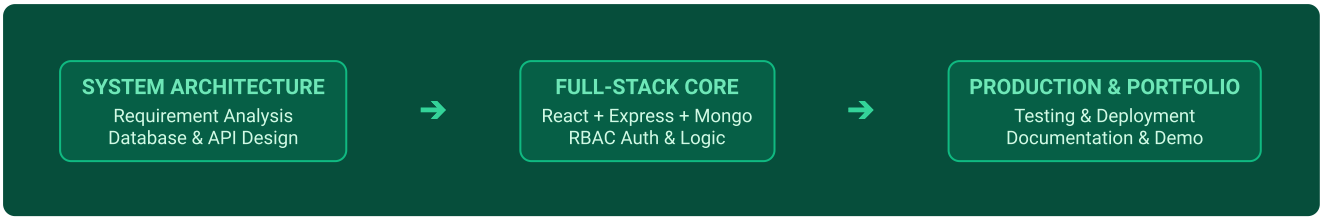
MONTHLY INDUSTRY SPECIFICATION

DURATION	TOTAL WORKLOAD	WEEKLY COMMITMENT	MODE & REVIEW
1 Month (4 Weeks)	40 Hours / Month	~10 Hours / Wk	Online + Mentor

1. Month 3 Overview & Monthly Goals

The objective of Month 3 is to shift interns from isolated technical learning to building production-grade, industry-level full-stack applications. Interns will analyze complex requirements, architect multi-tier systems, build business logic, implement security and testing, and deploy a complete project independently.

2. Industry-Grade Project Lifecycle



3. Comprehensive Monthly Tasks & Project Scope

Interns must complete the following integrated core modules to deliver their industry project package:

Task 1: System Architecture, Database Design & API Planning

- **Requirement Analysis:** Define problem statements, target user roles (e.g., Admin, User, Provider), and non-functional requirements.
- **Architecture & Schemas:** Design multi-entity database schemas with relationships and specify full REST API endpoints prior to coding.

Task 2: Full-Stack Core Development (React, Express & MongoDB)

- **Backend Engine:** Develop a modular Express server with controllers, models, routes, custom error middleware, and database validation.
- **Frontend Application:** Build component-driven React applications consuming backend APIs with state management and dynamic UI feedback.

Task 3: Role-Based Authentication & Advanced Business Logic

- **Authentication & RBAC:** Secure routes using `bcrypt` password hashing, JWT authorization, and role-based access control.
- **Domain Logic & Validation:** Implement complex feature workflows (e.g., booking, cart/checkout, tracking) alongside robust error handling.

Task 4: Production Deployment, Testing & Portfolio Deliverable

- **Testing & Security:** Conduct functional/API testing (Postman) and ensure `.env` key protection and zero plain-text secrets.
- **Deployment & Docs:** Deploy frontend (Vercel) and backend (Render/Railway), and assemble a professional GitHub portfolio with full README docs.

4. Overall Monthly Evaluation Criteria Breakdown (Total: 100%)

EVALUATION AREA	WEIGHT	TECHNICAL SCOPE & EVALUATION FOCUS
Project Planning & Architecture	**10%**	Requirement specs, system architecture diagrams, database schemas, and API endpoint design.
React & Frontend Implementation	**15%**	Component hierarchy, state management, controlled forms, router integration, and responsive UI.
Backend & API Development	**15%**	Express server structure, controllers, middleware, status codes, and modular route architecture.
Database & Data Management	**10%**	Mongoose model design, schema validation, entity relationships, and CRUD persistence.
Authentication & Security	**10%**	Password hashing (`bcrypt`), JWT implementation, protected routes, and Role-Based Access Control.
Business Features & Logic	**10%**	Domain-specific workflows, business logic implementation, and edge-case handling.
Testing & Debugging	**10%**	Postman API test collections, functional frontend/backend verification, and bug fixes.
Production Deployment	**5%**	Live deployment of frontend (Vercel) and backend (Render/Railway) connected to MongoDB Atlas.
Code Quality & GitHub Practice	**5%**	Repository organization, meaningful commit history, `.gitignore` usage, and clean code.
Documentation & Presentation	**5%**	Comprehensive `README.md`, architecture docs, screenshots, and technical explanation.
Independence & Problem Solving	**5%**	Ability to research, debug, and implement requirements without continuous spoon-feeding.

5. Security & Secrets Management Policy

Mandatory Security Directives

- **Secrets Isolation (.env):** Database credentials, JWT secrets, and private API keys must never be committed to GitHub repositories. Include `.env` in `.gitignore`.
- **Password Security:** All user passwords must be hashed using `bcrypt` prior to storage. Plain-text password storage is strictly forbidden.
- **Role Authorization:** Backend routes must enforce strict permission checks to prevent unauthorized access across user roles.

6. Final Submission Deliverables & Ethics Guidelines

Deliverables & Ethics Policy

At the conclusion of Month 3, interns must submit their full industry project package containing:

- **GitHub Repository:** Professional repository containing `/frontend` and `/backend` folders, clean commit history, and a detailed `README.md`.
- **Live Production Links:** Deployed live URLs for both the Frontend interface and Backend API service.
- **Documentation Package:** Architecture diagram, database schema blueprint, API endpoint list, and Postman testing collection.
- **Originality & Code Ownership:** Utilizing documentation and AI tools for debugging is permitted, but plagiarizing complete projects is strictly prohibited. Interns must be capable of independently explaining their technical implementation.